

Hybrid CNN–Transformer Networks with Clinical Metadata for Skin Lesion Classification

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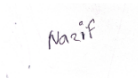
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Declaration

It is hereby declared that :

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Dermoscopy technique for classification of skin lesions may be useful in early diagnosis of severe skin diseases and in the clinical management of skin lesions. The challenge is to create an accurate and clinically usable system. We create two versions of this design, based on requirements. The first one is light in weight. It adopts a truncated MobileNetV2 backbone along with a compact transformer, and with only 3.98 million parameters, it achieves 60.8% BMA (balanced multi-class accuracy). This compact design facilitates easy installation within typical clinic environments, where space is a constraint. The second version emphasizes more balanced accuracy than capacity of the parameters. It is composed of two different pre-trained backbones (DenseNet-121 and EfficientNet-B0) and a learned attention mechanism that selectively fuses information from both followed with a Kolmogorov Arnold network as the final classifier and attains a BMA of 64.4% on 16.45M parameters. A dermatologist might consider the clinical context in the classification process but this information is seldom taken into account by image-based classifiers. In this case, both models include patient metadata as added diagnostic context directly into the classification pipeline with the data for age, sex and location of the body part where the lesion occurred. The models provide an efficient and accurate balance for real-world dermatology applications over same-budget baselines.

Keywords: Skin lesion classification; ISIC-2019; dermoscopy; CNN–Transformer; metadata fusion; cross-attention; Kolmogorov–Arnold Network; class imbalance; balanced multi-class accuracy; lightweight deep learning.

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Chapter 1

Introduction

1.1 Background

Skin cancer is among the most common cancers worldwide, and the prognosis for a patient depends heavily on how early the lesion is found. Melanoma in particular is highly treatable when caught early but becomes far more dangerous once it spreads. Dermoscopy a non-invasive imaging technique that removes surface glare and reveals the sub-surface structure of a lesion has become a standard tool for this kind of early assessment. Reading dermoscopic images well, however, takes years of training, and even experienced dermatologists do not always agree with one another. This gap between the volume of lesions that need screening and the number of specialists available to read them is what motivates automated, image-based diagnostic support.

In the last ten years, the field has been revolutionized by deep learning. CNNs trained on huge dermoscopic collections have achieved the dermatologist level performance in the recognition of melanoma, as the annual ISIC challenges have now established skin lesion classification as a well-defined benchmark task [5]. The ISIC-2019 dataset, which we use throughout this work, contains 25,331 labelled dermoscopic images spanning eight diagnostic categories: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BKL), dermatofibroma (DF), vascular lesion (VASC), and squamous cell carcinoma (SCC).

More recently, Vision Transformers (ViTs) [21] and hybrid designs that place a transformer on top of convolutional features [23], [24] have shown that modelling the global structure of a lesion its overall asymmetry, the regularity of its border complements the local texture that convolutions capture so well. These two views of a lesion, the local and the global, are exactly the cues a clinician is trained to weigh, which makes the CNN–Transformer combination a natural fit for dermoscopy.

1.2 Rationale of the Study or Motivation

Two practical problems keep most published skin-lesion models from being used in an everyday clinic. The first is cost. The strongest results on ISIC-2019 come from large ensembles of heavy networks [15], [19]; these are accurate but impractical to run on

the modest hardware found in a typical dermatology practice or on a portable device next to the patient. The second is that image-only models ignore information a dermatologist always has in front of them. The age of the patient, their sex, and the part of the body where the lesion sits all shift the prior probability of each diagnosis a scaly patch on the face of an elderly patient is read very differently from the same patch on a young person’s back. This clinical context is recorded in the ISIC-2019 metadata, yet it is routinely discarded by classifiers that look only at pixels.

Our study is driven by the belief that these two problems should be solved together rather than separately. We argue first that a carefully designed lightweight hybrid can match or beat much larger image-only baselines on balanced accuracy while staying small enough to deploy, and second that the clinical metadata a dermatologist already has on hand can be folded in at negligible cost as additional diagnostic context, handled so that missing fields are never imputed. We also recognise that not every deployment has the same constraints, so we deliver two models from one design philosophy: a genuinely compact model for constrained settings and a higher-capacity model for situations where balanced accuracy matters more than footprint.

1.3 Problem Statement

ISIC-2019 is severely imbalanced. The most common class, nevus, has roughly 12,875 images, while dermatofibroma and vascular lesions have only about 239 and 253 respectively a ratio of more than fifty to one. A model trained naively on this data learns to predict the common classes and quietly ignores the rare ones, which is the opposite of what matters clinically, since several of the rare classes are malignant. Reporting overall accuracy hides this failure entirely. The problem we address is therefore to build a model that performs well *across all eight classes at once* measured by balanced multi-class accuracy (BMA), the mean of the per-class recalls while respecting a tight parameter and compute budget and while making use of the clinical metadata that accompanies each image, including the large fraction of records where some of that metadata is missing.

1.4 Objective

The objectives of this work are:

1. To design a lightweight hybrid CNN Transformer classifier for ISIC-2019 that stays under 6M parameters and 1 GMAC and that fuses clinical metadata through a cross-attention head capable of handling missing fields without imputation.
2. To design a second, higher-capacity model in the same family a dual-backbone network with learned attention fusion and a Kolmogorov Arnold Network (KAN) classifier that trades footprint for a higher balanced accuracy.

3. To train both models, together with five established lightweight baselines, under one identical loss, augmentation, and evaluation pipeline so that the comparison is fair.
4. To quantify the contribution of each design choice (the CNN stem, the transformer trunk, the metadata head, and the dual-backbone fusion) through controlled ablations, and to report efficiency alongside accuracy.

1.5 Methodology in Brief

Every image passes through a fixed preprocessing pipeline: near-duplicate removal by perceptual hashing, a lesion-grouped stratified train/validation/test split, Shades-of-Gray colour constancy [1], and resizing to 224×224 . The cleaned images are stored as a single memory-mapped array so that training does not pay a per-batch decoding cost. All models are trained with Class-Balanced Focal Loss [7], [14] to counter the imbalance, with AdamW, a cosine schedule with warmup, weight EMA, and eight-view test-time augmentation at inference. The lightweight model combines a truncated MobileNetV2 stem with a small ViT trunk and a metadata cross-attention head; the flagship model (DEKAN) fuses DenseNet-121 and EfficientNet-B0 features through learned attention before a transformer trunk and a KAN classifier. The primary metric throughout is BMA.

1.6 Scopes and Challenges

The scope of this thesis is the eight-class ISIC-2019 classification task and the two proposed architectures, evaluated against same-budget lightweight baselines on a held-out test split. The main challenges were the extreme class imbalance, the substantial missingness in the metadata (around 30% of age values, for example) and a hard hardware budget all experiments run on a single 16 GB GPU under Windows, which rules out multi-worker data loading and forces aggressive use of mixed precision and a memory-mapped dataset. The study was not extended to external test sets nor to multi-seed reporting which are both discussed openly as limitations in Chapter 5.

Chapter 2

Literature Review

2.1 Preliminaries

It is helpful to establish a few ideas in the mind that repeat during the reading of previous work before beginning to read it. A convolutional neural network builds up a representation of an image through stacked local filters, which makes it very good at capturing texture and short-range structure but, by construction, slow to relate distant parts of an image to one another. A transformer takes the opposite stance: it treats the image as a set of tokens and lets every token attend to every other through self-attention [9], [21], so global relationships are available from the first layer at the cost of weaker built-in spatial priors and a hunger for data. The *attention* operation at the heart of both transformers and our fusion modules computes a weighted average of “value” vectors, where the weights come from the similarity between a “query” and a set of “keys”; when the query comes from one source and the keys and values from another, the same mechanism becomes *cross-attention*, which is how we inject clinical metadata into the image representation. Finally, a Kolmogorov–Arnold Network (KAN) [25] replaces the fixed activation and linear weights of an ordinary layer with learnable spline functions on the edges of the network, a recent idea we evaluate as a classifier head.

2.2 Review of Existing Research

2.2.1 Convolutional backbones used in this work

The convolutional components of both our models are standard, well-understood backbones, chosen because their behaviour and pretrained weights are reliable. **MobileNetV2** [10] introduced the inverted residual block with a linear bottleneck, which keeps the network cheap by expanding to a higher-dimensional space only inside each block and using depthwise convolutions. It has a very efficient way of encoding generic edges and textures at the early stages; this is why we use a truncated MobileNetV2 as the stem of our lightweight model. **EfficientNet-B0** [17] is a base model of a family designed from compound scaling of its depth, width, and

resolution properties, which is around 5M parameters and is a good accuracy-per-FLOP baseline, and one of the two backbones in our flagship model. The second backbone is **DenseNet-121** [6] which invokes the reuse of features and provides a rich representation that is gradient friendly; it is implemented in the flagship model. We also consider the residual-learning network, **ResNet-18** [2] as an easy-to-follow example.

2.2.2 Transformers and CNN Transformer hybrids

By dividing an image into patches and regard them as a token sequence, the **Vision Transformer** [21] demonstrated that a pure transformer, with sufficient data, can perform as well as a convolutional model for image classification tasks. Its limited effectiveness on compact data sets led to hybrid designs which preserve a convolutional front end. **MobileViT** [24] interleaves convolutions with transformer blocks to get a mobile-friendly model that still reasons globally, and **EfficientFormer** [23] shows that transformer-grade accuracy can be reached at MobileNet-like latency with a carefully chosen dimension-consistent design. We treat both as direct, same-spirit baselines, since a reviewer will reasonably ask how a new hybrid compares against existing lightweight hybrids rather than against pure CNNs alone. Our own design differs from these in where the convolution-to-transformer hand-off happens and, more importantly, in that the transformer’s class token is used as a query that attends over clinical metadata.

2.2.3 Kolmogorov Arnold Networks

KANs [25] are a recent alternative to the multilayer perceptron, grounded in the Kolmogorov Arnold representation theorem, in which the learnable parameters are the coefficients of B-spline functions placed on the network’s edges rather than scalar weights feeding fixed activations. The promise is more expressive, more interpretable units, sometimes at fewer parameters. KANs are still new and most evidence comes from small scientific-regression problems, so whether they help as a classifier head on a noisy, imbalanced vision task is an open question. We include a KAN head as the final classifier in our flagship model to explore whether its learnable splines add value at the output stage of a well-trained vision backbone.

2.2.4 Clinical metadata fusion

The work closest in spirit to ours is by Pacheco and Krohling [20], [22]. They showed first that adding patient clinical information measurably improves automated skin cancer detection [20], and then proposed an attention-based mechanism to combine images and metadata inside a deep model [22]. Gessert et al. [19], the basis of the top ISIC-2019 challenge entry, also fed metadata into an ensemble of EfficientNets. The key methodological point we adopt from this line of work is that missing metadata should be represented explicitly with a learned “missing” embedding rather than imputed with a fabricated value, so the model receives an honest “unknown” signal. Where we depart from prior work is in the backbone: Pacheco and Krohling fused

metadata onto heavy CNNs, whereas we attach the same idea to a lightweight hybrid and to a dual-backbone hybrid, and we make an explicit efficiency claim about the result.

2.2.5 Skin lesion classification and ISIC benchmarks

Beyond the specific components above, automated dermoscopy has a large literature. Esteva et al. [5] demonstrated dermatologist-level melanoma classification with a single large CNN, and the ISIC challenges [4] have since standardised evaluation. The ISIC-2019 dataset is itself a union of several sources, most prominently HAM10000 [11] and BCN20000 [13], which is why source-aware splitting matters when building a fair evaluation. On the ISIC-2019 leaderboard [15], where the official ranking metric is balanced accuracy, the strongest published entries are large ensembles: the top entry (DAISYLab) reached a balanced accuracy of 0.636 with an ensemble of multi-resolution EfficientNets plus SENet-154, followed by entries around 0.61 and 0.59, again ensembles of EfficientNet and SE-ResNeXt variants. These numbers provide an informative cap and a reminder that the efficiency ranges of single, lightweight models are different than the efficiency ranges of the leaderboard ensembles. Most relevant to our experimental design, recently Aktan et al. [26] evaluated several popular deep learning architectures on this same dataset in the same manner under a single protocol, giving a clean, single-model reference of the behaviour of standard backbones on ISIC-2019 under this protocol.

2.3 Summary of Key Findings

Three observations from the literature shape our design. First, hybrid CNN–Transformer models are a sensible match for dermoscopy because lesions are defined by both local texture and global shape, but the existing lightweight hybrids do not use clinical context. Second, metadata fusion clearly helps on this task, and the practical detail that decides whether it helps is how missing values are handled learned embeddings rather than imputation. Third, the best-published numbers on ISIC-2019 come from heavy ensembles, which leaves an open and useful niche for a single, small model that uses metadata to close part of the gap. This is precisely the gap the two models in this thesis are designed to fill.

Chapter 3

Requirements, Impacts and Constraints

3.1 Dataset and Data Requirements

All experiments use the ISIC-2019 training collection, a public set of 25,331 dermoscopic images labelled across the eight diagnostic classes listed in Chapter 1. The collection is itself a union of three sources HAM10000, BCN20000, and MSK which we infer from the image-ID prefix, because keeping track of the source matters for building an honest train/test split. Each image also carries clinical metadata: approximate age, sex, the general anatomical site of the lesion, and a lesion identifier that groups multiple images of the same physical lesion.

The defining property of this dataset, and the one that drives most of our design decisions, is its class imbalance. Table 3.1 shows the per-class counts for the full collection. The largest class, nevus, is more than fifty times larger than the smallest, dermatofibroma. Several of the rare classes (for example SCC and AK) are clinically important, so a model that simply learns the head of this distribution would be useless in practice. This is the central reason we report balanced multi-class accuracy rather than overall accuracy throughout.

Table 3.1: Class distribution of the full ISIC-2019 collection (before deduplication) and imbalance relative to the largest class (NV).

Code	Full name	Images	Ratio vs. NV
MEL	Melanoma	4,522	$2.8\times$
NV	Melanocytic nevus	12,875	$1\times$
BCC	Basal cell carcinoma	3,323	$3.9\times$
AK	Actinic keratosis	867	$14.8\times$
BKL	Benign keratosis	2,624	$4.9\times$
DF	Dermatofibroma	239	$53.9\times$
VASC	Vascular lesion	253	$50.9\times$
SCC	Squamous cell carcinoma	628	$20.5\times$

The metadata is genuinely incomplete: roughly 30% of age values, about 20% of sex values, and about 15% of anatomical-site values are missing, with the recorded ages centred near 54 years. Rather than fill these gaps with guessed values, our requirement is that the model treat “missing” as a first-class, learnable signal, which is reflected directly in the architecture (Chapter 4).

3.1.1 Functional Requirements

The functional requirements specify what the system must do, independent of how well it does it. They follow directly from the goal of a clinically usable, metadata-aware classifier and are summarised in Table 3.2.

The system must accept a single dermoscopic image and return a probability distribution over the eight ISIC-2019 diagnostic categories (MEL, NV, BCC, AK, BKL, DF, VASC, SCC), rather than a single hard label, so that the output is a ranked set of possibilities suitable for decision support. It must optionally accept the patient metadata that accompanies each image age, sex, and the general anatomical site of the lesion and fold that context into the prediction. Crucially, it must continue to function when any of those metadata fields is absent, representing “missing” as an explicit, learnable signal rather than imputing a fabricated value, because a large fraction of ISIC-2019 records are incomplete. The system must apply a fixed preprocessing pipeline (colour constancy and resizing to 224×224) to raw input before inference, and must report performance per class and as balanced multi-class accuracy (BMA) so that the behaviour on rare, often malignant, classes is always visible. Finally, the whole training and evaluation procedure must be reproducible from configuration files.

Table 3.2: Functional requirements of the proposed system.

ID	Requirement
FR1	Accept an RGB dermoscopic image and output a probability distribution over the eight ISIC-2019 classes.
FR2	Optionally accept patient metadata (age, sex, anatomical site) and use it as additional diagnostic context.
FR3	Handle any missing metadata field through a learned “missing” representation, without imputation.
FR4	Return a ranked list of candidate diagnoses with associated scores (decision support, not a final diagnosis).
FR5	Apply the full preprocessing pipeline (Shades-of-Gray colour constancy, resize/centre-crop to 224×224) to raw input.
FR6	Train, validate, and evaluate from declarative configuration files with a fixed random seed.
FR7	Report per-class recall, BMA, macro-F1, macro-AUC, overall accuracy, and a normalised confusion matrix.

3.1.2 Non-Functional Requirements

The non-functional requirements specify the quality attributes the system must satisfy the constraints on *how* it performs its function. They are the requirements against which the design in Chapter 4 is judged, and they are listed in Table 3.3.

The central non-functional requirement is efficiency: the lightweight model must stay under 6M parameters and 1 GMAC so that it can be deployed on the modest hardware found in a typical clinic or on a portable device. The delivered lightweight hybrid meets this comfortably at 3.98M parameters and 0.631 GMACs. Closely related is deployability the system must train and run within a 16 GB memory ceiling and portability, since the implementation must run on a standard PyTorch stack, including under Windows, where the data loader is restricted to a single worker process. On the quality-of-output side, the most critical measure of performance is balanced multi-class accuracy, which is aimed for by a model that is not likely to suffer the phenomenon of rare-class collapse commonly found in a naively trained classifier when applied to this extremely imbalanced dataset. The figures are to be regenerated and the pipelines deterministic; the codebase is to be modular, and split into preprocessing, datasets, losses, models, training and analysis so that every reported number can be traced to its source; the codebase is to be made reproducible (fixed seed, deterministic pipeline, regenerable figures). There is another set of ethical requirements: the use of BMA is an ethical requirement because it requires that the rare classes of the malignant group be counted as equal; the prohibition on imputing missing clinical metadata is an ethical requirement because it ensures that the model is never provided with a fake clinical value.

Table 3.3: Non-functional requirements and how the design satisfies them.

ID	Requirement	How it is met
NFR1	Efficiency / footprint: lightweight model $\leq 6\text{M}$ parameters and ≤ 1 GMAC.	Lightweight hybrid: 3.98M params, 0.631 GMACs.
NFR2	Deployability: train and run within a 16 GB memory budget.	AMP, channels-last layout, gradient accumulation, memory-mapped dataset.
NFR3	Accuracy: balanced multi-class accuracy as the primary metric; beat same-budget baselines.	Hybrid 0.6081 BMA (best in sub-6M budget); DEKAN 0.6438.
NFR4	Robustness to class imbalance: no rare-class collapse.	Class-Balanced Focal Loss, weighted sampler; stable training curves (Ch. 5).
NFR5	Reproducibility: deterministic, seed-fixed, configuration-driven.	Single seed (42); figures regenerated from saved outputs (Appendix A).
NFR6	Maintainability: modular, importable codebase.	Separate preprocessing / datasets / losses / models / training / analysis modules.
NFR7	Portability: standard framework; runs under Windows with zero-worker loading.	PyTorch + <code>timm</code> ; memmap dataset compensates for single-worker constraint.
NFR8	Fairness and honesty: rare malignant classes weighted equally; no fabricated metadata.	BMA metric; explicit learned “missing” embeddings.
NFR9	Latency: single-image inference fast enough for interactive use.	Low GMAC count (0.631) keeps per-image inference inexpensive.

3.2 Societal Impact

The purpose of this work is to provide a small, user friendly decision support tool, that exploits the same contextual information that a clinician has. Such a tool has implications on many fronts, including structural access to care, health and safety issues, legal considerations, and cultural considerations.

From a structural perspective, the increase in the number of skin lesions that need to be screened and the lack of dermatologists to examine them creates a big disconnect. The lack is particularly pronounced in low-resource countries, and in Bangladesh, specialist dermatological coverage is limited to certain major urban centers, with limited early screening coverage in rural areas. An inexpensive, offline-friendly model extends coverage and improves access to health care, especially where that coverage is limited. An inexpensive, offline-friendly model extends coverage and improves access to health care, not further centralizing it. Its realistic social role is to be a triage alerting system to lesions that need medical attention, as the system is assistive and does not take the place of the specialist judgment that is still in short supply.

The clinical advantage of this triage function arises from the prognosis of skin cancer

and the significant role of early detection in this disease, specifically early detection of melanoma can be very beneficial. Adding such a tool to a healthcare workflow, however, raises some critical safety issues. The worst failure case for a skin-lesion classifier is a false negative for malignant. This is why we report balanced multi-class accuracy instead of overall accuracy – a model that was highly accurate on the benign class of a nevus, but failed to correctly classify rare malignant cases would be impressive in terms of its accuracy, but unhelpful in the real world. The second-line safety concern is automation bias: a tendency of users to follow a trustworthy machine output. We minimise this by making it a ranked set of possibilities rather than a single diagnosis, and by making it make clear that it’s an aid to a clinician sitting in the loop for all decisions.

From a regulatory and legal standpoint, any clinical deployment of this system would sit within a medical-device regulatory regime and require corresponding clearance before use on patients; the present work makes no claim to such clearance. Liability for a missed or incorrect diagnosis, the need for informed consent governing the use of patient data, and compliance with data-protection law are all legal questions that a real deployment must answer. By relying only on a public, de-identified research dataset and by framing the output as decision support, this study stays within the bounds of research use and leaves these clinical-deployment questions explicitly out of scope.

Finally, the success and equity of this tool are deeply tied to cultural and demographic context. A specific and important limitation of this study concerns skin-tone representation. ISIC-2019 is assembled largely from European populations, and dermoscopic collections in general over-represent fair skin. A model trained on such data cannot be assumed to generalise to the darker and more diverse skin tones of a South Asian population such as Bangladesh’s. Deploying it there without re-validation on representative local data would risk systematically worse performance for the very patients it is meant to serve. We flag this as a genuine equity concern rather than a solved problem, and note it again among the limitations and future work as a requirement for external validation on a local, representative dataset. More broadly, the acceptance of AI-assisted diagnosis depends on cultural attitudes toward automation in healthcare and on patient trust, which reinforces the case for a transparent, assistive design rather than an opaque, autonomous one.

3.3 Environmental Impact

The environmental footprint of a deep learning project is best understood through two distinct phases: the one-off energy cost of training the models, and the recurring, compounding cost of running inference once deployed. Efficiency is a usually interpreted as a deployment benefit of machine learning, but actually it is a sustainability argument. Reduced computational resources can directly be translated into reduced consumption of energy and a lower carbon footprint of the software during its lifecycle.

We measured the initial environmental impact by examining the impact of our training phase. All experiments are performed on one machine with an NVIDIA

RTX 4070 Ti Super GPU, with a 285 W board capacity and an estimated average consumption of 0.25 kW during the normal training. The carbon cost of a single (one) training run is:

$$\text{CO}_2\text{eq} = P_{\text{avg}} \times t_{\text{train}} \times \text{EF},$$

P_{avg} is the average board power, t_{train} is the wall clock training time and EF is the grid emission factor. The official grid emission factor of 0.62 kg CO₂/kWh is relatively high for electrical grid because of Bangladesh’s dependence on fossil fuel.

We will use the wall- clock times from our training logs in our two proposed models, and apply this framework to them in Table 3.4. When scaling this math across the entire project—which included five baselines, both proposed architectures, and various ablation variants totaling roughly a dozen full training runs alongside minor exploratory sessions—the total training energy falls on the order of tens of kWh. This amounts to a few tens of kilograms of CO₂eq, a total footprint roughly equivalent to a single short drive in a gasoline-powered car. Crucially, this is orders of magnitude lower than the massive carbon expenditure required to train the heavy model ensembles that typically dominate the top of the ISIC-2019 leaderboard.

Table 3.4: Estimated training energy and carbon for the two proposed models, assuming an average board power of 0.25 kW and a grid emission factor of 0.62 kg CO₂/kWh.

Model	Wall-clock (h)	Energy (kWh)	CO ₂ eq (kg)
Lightweight hybrid	≈ 3	0.75	0.47
DEKAN (flagship)	≈ 9	2.25	1.40

While training is a sunk cost, the recurring inference cost dominates a clinical tool’s lifetime footprint once it is deployed in the field. This is where our architectural design choices offer the greatest sustainability dividends. Because the lightweight hybrid processes a single classification using only 0.631 GMACs, its energy consumption per inference is practically negligible. This compact profile allows the model to run locally on low-power edge devices, such as an NVIDIA Jetson Orin Nano, which handles vision-transformer workloads within a modest 7–25 W power envelope. This bypasses the need for a continuously active, server-class GPU that draws ten to twenty times more power.

In addition, a compact model contributes to sustainability well beyond its electricity use: since it can be hosted by inexpensive or existing equipment, it helps to keep costs down. This decreases the demand for new top-dollar accelerators in institutions and helps resolve the rising tide of e-waste. It also eliminates the need for a persistent data center connection with high bandwidth. In contrast, the now state-of-the-art multi-resolution EfficientNet and SENet-154 ensembles require an enormous computational cost for each image in the target set. To make the choice between efficiency and accuracy accessible, this study introduces a two-tier design, reflecting the tenets of green machine learning with a low footprint path that balances clinical utility with efficiency.

It needs to be recognized that some drawbacks exist in these measurements. No power measurements were taken on the electricity meter at the wall socket, but

these are based on the rated power of the GPU and average electricity data from the national grid. Actual values are expected to vary in real-world situations, depending on exact board loading and variations in the energy mix. These figures are given as a good order of magnitude estimate, and it is enough to prove these models are at least one order of magnitude lower on the environmental cost compared to the standard leaderboard architecture.

3.4 Ethical Issues

This ethical aspect of the work also shaped the technical design, such as when considering data privacy, fairness of algorithms and transparency when reporting to the clinic.

In terms of data ethics and privacy, this work only uses the public ISIC- 2019 dataset provided by the International Skin Imaging Collaboration. No additional human subject data was gathered and all images were anonymized at the source. Only limited patient information like age, sex and anatomical site is used as input for the model to enhance the diagnostic support. This information is used with strict compliance to the terms of use of ISIC data and does not risk the re-identification of patients.

There are two main technical decisions to implement algorithmic fairness. Optimizing for balanced multi-class accuracy forces the model to give equal importance to rare, potentially malignant classes, rather than the majority benign classes; this is because the majority benign classes would otherwise dominate the performance measures. Optimizing for balanced multi-class accuracy ensures that the model treats rare classes, potentially malignant classes, as important as the majority benign classes; otherwise, the majority benign classes would be dominating the results. Second, missing metadata is handled via a learned “missing” embedding rather than statistical imputation, ensuring the model never trains on fabricated clinical values. However, a significant fairness limitation remains: the skin-tone representation bias inherent in the ISIC-2019 dataset, which is largely derived from fair-skinned European populations. This remains an unresolved equity issue that requires external validation on local, diverse populations before any real-world deployment.

Finally, to ensure responsible deployment, the system is explicitly designed as a collaborative decision-support tool rather than an autonomous diagnostic agent. By outputting a ranked list of differential possibilities, the architecture keeps a clinician firmly in the loop and mitigates automation bias. In the interest of scientific transparency—which is a safety obligation in medical AI we report our negative and inconclusive results with equal weight to our successes. Specifically, we openly disclose that the metadata benefit for the lightweight model remained unconfirmed on a single test-set evaluation and that the KAN classifier head yielded no performance gains over a standard linear head.

3.5 Project Management Plan

This project was executed by a team of five over three academic semesters, spanning from Spring 2025 to Summer 2026, under the supervision of Dr. Md. Ashraful Alam. The work was structured into three sequential phases—Pre-Thesis 1, Pre-Thesis 2 and the Final Thesis with specific tasks mapped to the timeline shown in Table 3.5.

The first part covered basic research and data engineering, such as a thorough literature review, a problem formulation, on the data imbalance issue, and the development of a comprehensive preprocessing pipeline (comprising perceptual-hash deduplication, lesion-grouped stratified splitting, color constancy, and a memory-mapped data set). The second phase started the process of model development, putting in place five lightweight baselines with the same training recipe and designing a lightweight hybrid model. The final stage was the development and training of the higher capacity DEKAN architecture, the execution of controlled ablation studies, a careful analysis of the architecture’s performance, and the writing of the final thesis manuscript.

Table 3.5: Project schedule across the three thesis phases. A filled cell (■) indicates active development during that phase.

Stage	Pre-Thesis 1	Pre-Thesis 2	Final Thesis
Problem definition & literature review	■	□	□
Data acquisition & preprocessing pipeline	■	■	□
Baseline implementation (5 models)	□	■	□
Lightweight hybrid design & training	□	■	■
DEKAN design & training	□	□	■
Ablation studies	□	□	■
Evaluation, analysis & figures	□	■	■
Thesis writing & revision	■	■	■

Responsibilities were structured in parallel with the technical pipeline for accountability and to meet individual contribution criteria. The team was kept synchronized with strict Git version control and weekly progress checks with the supervisor.

In terms of resources and infrastructure, all training consisted of a single common workstation (detailed in Section 3.1.4). Managing the entire project on a single machine within a configuration-driven pipeline was a design decision, as such, it provided complete control of the environment drift, and allowed for a true apples-to-apples comparison across all 5 models. Important project deliverables accomplished include a complete deduplicated and stratified dataset, a verified baseline suite, a lightweight hybrid model at an affordable cost, the flagship DEKAN model, an ablated matrix completed, and the last appendix of reproducible code.

3.6 Risk Management

During the initial stages of the project, a few technical problems were identified as threats to validity. These risks were considered and addressed explicitly within the experimental design, and were a deliberate choice in the engineering design. Each risk identified is recorded on the risk register in Table 3.6, along with the pre-mitigation risk severity, the engineering solution used and the risk status after the solution was applied.

Leakage of data was the most pernicious cause. The ISIC-2019 dataset is composed of many duplicate images of the same lesion, making it impossible to naively split the images into training and test sets without including the same lesions in both, which would bias the results. We eliminated this risk by implementing perceptual-hash (pHash) deduplication followed by a strict lesion-grouped, source-stratified split. Another major threat was rare-class collapse, where severe class imbalance could cause a model to maximize overall accuracy by simply ignoring rare malignant conditions. This was successfully countered by employing a Class-Balanced Focal Loss, utilizing a weighted sampler, and choosing Balanced Multi-class Accuracy (BMA) as our primary selection metric; training logs confirm that no such collapse occurred.

Table 3.6: Project risk register (L = Likelihood, I = Impact, graded as High/Medium/Low).

Risk	L	I	Mitigation	Status
Train/test leakage via duplicate lesion images	H	H	pHash deduplication + lesion-grouped, source-stratified split	Mitigated
Rare-class collapse under imbalance	H	H	Class-Balanced Focal Loss, weighted sampler, BMA selection metric	Mitigated
Overfitting of from-scratch transformer on $\sim 17k$ images	M	M	Pretrained CNN stem, stochastic depth, RandAugment, Mixup, weight EMA	Mitigated
16 GB GPU memory ceiling	H	M	AMP, channels-last, gradient accumulation, memmap dataset	Mitigated
Windows data-loader deadlock	M	M	Zero-worker loading + memory-mapped array	Mitigated
Substantial metadata missingness	H	M	Learned “missing” embeddings, no imputation	Mitigated
Single-seed variance	H	M	Acknowledged; multi-seed reporting deferred	Open (future work)
No external-dataset validation	M	H	Acknowledged; external validation deferred	Open (future work)
KAN head failing to add value	M	L	Linear-head ablation retained and reported	Realised, handled

We also encountered strict hardware and environmental constraints, namely a 16 GB GPU memory ceiling and a known Windows OS data-loader deadlock. These infrastructure risks were circumvented by adopting Automatic Mixed Precision (AMP), a channels-last memory layout, gradient accumulation, and a memory-mapped, zero-worker dataset configuration that bypassed the OS multi-processing limitation entirely. Similarly, the risk of training on fabricated clinical values due to missing patient metadata was resolved by introducing learned “missing” embeddings rather than statistical imputation.

Two risks could not be fully resolved within the current scope and are carried for-

ward as acknowledged limitations: all reported results are derived from a single experimental seed, and the architectures have not yet been evaluated on an independent external dataset. Consequently, true seed variance and out-of-distribution generalization remain open questions for future validation. Finally, the risk of the KAN classifier head failing to add value was realized during testing; however, this was handled transparently by maintaining and reporting the linear-head ablation variant rather than suppressing the negative result.

3.7 Economic Analysis

This work is not so costly to develop as to be unaffordable, but rather more expensive in terms of ongoing operating costs. A major financial benefit over cloud-hosted inference infrastructures and the large model ensembles that usually dominate the research leaderboard is achieved by designing a compact model.

The development expenses are mostly independent of the number of developers utilised: a single workstation (Section 3.1.4) that costs around USD 1,800 to 2,500, together with the research team’s time over a period of three academic semesters. The software package is 100(%) open source, and free. In addition, the electricity requirement for model training is very low since the total electricity consumption of the study was only tens of kWh as computed in Section 3.3. This would just cost a few dollars at a typical commercial electricity price of about USD 0.08/kWh in Bangladesh.

The real economic disparity is at the ‘deployment’ stage. The lightweight hybrid model is made with a mere 3.98M parameters and 0.631 GMACs and hence can be served locally from a cheap edge hardware. A dedicated device, such as the NVIDIA Jetson Orin Nano Super, specifically designed for vision-transformer workloads, costs around USD 249 once up and running, consumes just a few watts of power, and requires no internet connection or WiFi. This offline functionality gives huge operational edge to the rural clinics where network connectivity is weak.

On the other hand, cloud-based GPU inference has significant recurring operating expenses. Even a small T4-class instance has an on-demand fee of 0.15–0.81 per GPU-hour. Having an endpoint that’s always available will cost hundreds of dollars monthly, and don’t forget that data egress fees and complicated legal/privacy issues arise when patient images are sent offsite. The scenario for clinical use in Table 3.7 assumes a three-year Total Cost of Ownership (TCO) comparison, based on a plausible clinical use case. Overall, the edge architecture is about an order of magnitude cheaper over its entire lifetime when compared with leaderboard ensembles that need enterprise-grade data center accelerators.

Table 3.7: Illustrative three-year Total Cost of Ownership (TCO) for model deployment. Cloud estimates assume a T4 instance at USD 0.50/GPU-hour active 8 hours per day; edge estimates reflect the hardware purchase price plus local electricity tariffs.

Option	Up-front (USD)	3-year running (USD)	3-year total (USD)
Edge (Jetson Orin Nano Super, ~10 W)	249	~9	~258
Cloud (T4-class, 8 h/day on demand)	0	~4,380	~4,380

This dramatic cost disparity is the basis for the project’s cost-benefit argument. The cost per patient of a USD 249 device over thousands of screenings is a low cost structural societal value for early lesion triage. Early detection of one malignant lesion results in an enormous amount of economic and clinical savings because late stage cancer therapies are not used.

The narrow accuracy gap between our lightweight model (0.6081 BMA) and a heavy leaderboard ensemble (≈ 0.636) is easily justified by the order-of-magnitude reduction in deployment barriers. For clinical environments where maximizing diagnostic accuracy warrants a larger operational budget, the flagship DEKAN model (0.6438 BMA) offers a higher-performance alternative within the same design paradigm. This architecture ultimately provides healthcare institutions with a flexible, principled choice between two distinct cost-performance tiers rather than forcing a one-size-fits-all deployment strategy.

Chapter 4

Proposed Methodology

This chapter describes the full pipeline: how images and metadata are prepared, the two proposed architectures, the loss that handles the class imbalance, and the training and evaluation protocol shared by every model in the study.

4.1 Preprocessing Pipeline

Raw dermoscopic images vary in size, illumination, and — because ISIC-2019 is a union of several archives — in the presence of near-duplicate photographs of the same lesion. The pipeline cleans all of this once, before any training, in three steps.

Near-duplicate removal. We compute a 64-bit perceptual hash (pHash) for every image and group images whose hashes are within a Hamming distance of 4 using union-find, keeping a single representative per group. This removed 1,263 near-duplicates, leaving 24,068 unique images. Removing duplicates before splitting is what prevents the same lesion from leaking between training and test.

Lesion-grouped stratified split. We split the data 70/10/20 into training, validation, and test sets. The split is stratified jointly on class and source (HAM/BCN/MSK) so each split has a representative mix, and it is grouped by lesion identifier so that every image of a given physical lesion lands in the same split. The result is 16,872 training, 2,400 validation, and 4,796 test images.

Colour constancy and memmap. Each kept image is corrected with Shades-of-Gray colour constancy [1] (Minkowski norm $p = 6$) to reduce the strong illumination differences between dermatoscopes, then short-side-resized and centre-cropped to 224×224 . The entire cleaned dataset is written once to a single memory-mapped `uint8` array together with the labels and metadata. At training time this array is opened read-only and stays resident in the host’s page cache, so after the first epoch a batch costs little more than a memory copy — the key optimisation that makes single-worker data loading on Windows fast enough.

4.2 Overview of the Proposed Models

We propose two models that share a common idea — convolutional features, a transformer trunk for global reasoning, and a cross-attention head that folds in clinical metadata — but sit at two different points on the accuracy/efficiency curve. Figure 4.1 sketches both. The first, which we refer to as the *lightweight hybrid*, is built to stay under 6M parameters. The second, *DEKAN*, spends more capacity in exchange for higher balanced accuracy.

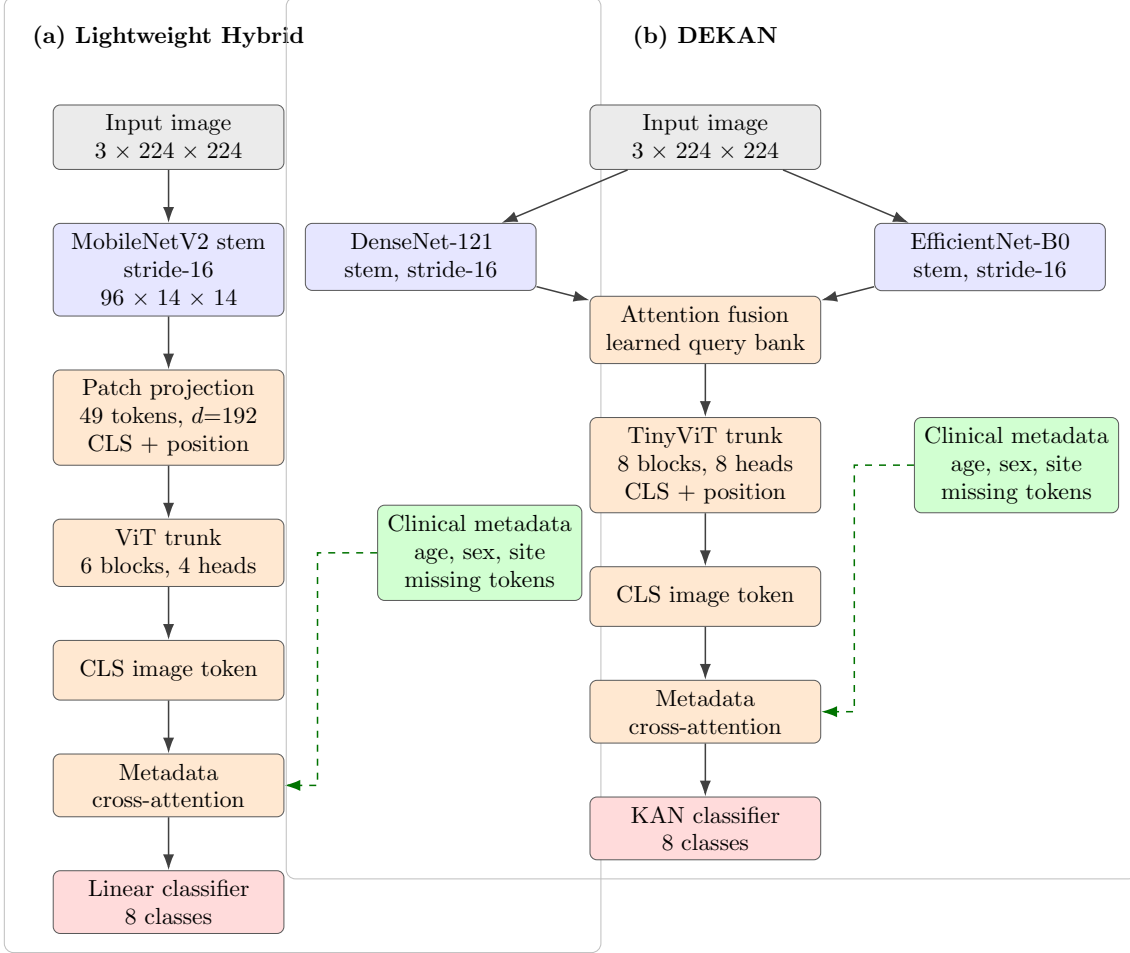


Figure 4.1: High-level architecture of the two proposed models. Both share the same philosophy convolutional features, a transformer trunk, and a metadata cross-attention head at two capacity points. (a) the lightweight hybrid uses a single truncated MobileNetV2 stem and a 6-layer ViT trunk with a linear head; (b) DEKAN fuses two pretrained backbones (DenseNet-121, EfficientNet-B0) through a learned attention bank, runs an 8-layer TinyViT trunk, and ends in a KAN head.

4.3 Lightweight Hybrid CNN–Transformer

4.3.1 CNN stem

The stem is a MobileNetV2 [10] (width 1.0), loaded with ImageNet weights and truncated at the stride-16 stage so that a 224×224 image produces a $96 \times 14 \times 14$ feature map. Using a pretrained convolutional front end gives the model generic edge and texture detectors for free, which matters because the transformer that follows is trained from scratch on only about 17,000 images. In our budget accounting the stem accounts for roughly 1.81M of the model’s parameters.

4.3.2 Transformer trunk

A 2×2 stride-2 convolution projects the 96-channel, 14×14 feature map into a sequence of $7 \times 7 = 49$ tokens of dimension 192; this both embeds the patches and halves the spatial resolution so that self-attention, which is quadratic in the number of tokens, stays cheap. A learnable class (CLS) token is prepended and a learnable positional embedding added, giving 50 tokens. These pass through six pre-norm transformer blocks with four attention heads, an MLP expansion ratio of 2, and a linearly increasing stochastic-depth rate [3] up to 0.1, which is what keeps a small from-scratch transformer from overfitting at this data scale. The CLS token after the final layer summarises the image globally.

4.3.3 Metadata cross-attention head

The clinical metadata is fed into a single cross-attention block with the image CLS token used as a query, and three metadata tokens (age, sex, anatomical site) as keys and values. A linear projection is used for age - a continuous variable, while sex and site - a categorical variable - are mapped to a token by using learned embeddings. Explicit handling of missing values: age has a separate learned “missing” vector that is used instead of the projection when age is missing; the categorical embeddings use an additional learned “missing” vector when the category is missing. There is also a learned type embedding embedded in each token, which enables the attention to distinguish the three fields. The block lets the model nudge the image representation using context — for instance, raising the prior on actinic keratosis for an older patient — without ever being forced to act on a fabricated value. The updated CLS token feeds a linear classifier producing eight logits.

The complete lightweight hybrid has 3.98M parameters and 0.631 GMACs, comfortably inside the budget of 6M parameters and 1 GMAC.

4.4 DEKAN: Dual-Backbone Flagship Model

DEKAN keeps the same overall shape but invests capacity where it buys accuracy.

Two complementary backbones. Instead of one stem it runs two in parallel — DenseNet-121 [6] and EfficientNet-B0 [17], both ImageNet-pretrained and both truncated to the stride-16 stage. The two networks encode the same lesion through different inductive biases, and the model is given the chance to use whichever is more informative per region.

Learned attention fusion. Each backbone’s feature map is projected to 49 tokens of dimension 256 and tagged with a per-backbone embedding. A learnable bank of 49 query tokens then cross-attends over the concatenation of both token sets (98 tokens) and produces a single fused sequence of 49 tokens. This is a learned, content-dependent fusion rather than a fixed concatenation or average: the query bank decides, position by position, how much to take from each backbone.

Transformer trunk and metadata. A CLS token and positional embedding are prepended to the fused sequence, which then passes through an eight-layer TinyViT-style transformer trunk (dimension 256, eight heads, MLP ratio 4), followed by the same metadata cross-attention head described above.

KAN classifier. The final classifier is a single Kolmogorov–Arnold layer [25] mapping the 256-dimensional CLS representation to eight logits, with a grid size of 5 and cubic (order = 3) B-splines. Its spline basis is computed in full precision for numerical stability even under mixed precision. We deliberately keep a linear-head version of the otherwise-identical model so that the KAN’s contribution can be measured directly (Chapter 5).

DEKAN has 16.45M parameters and 6.633 GMACs.

4.5 Class-Balanced Focal Loss

To counter the imbalance in Table 3.1, every model is trained with Class-Balanced Focal Loss, which combines the effective-number reweighting of Cui et al. [14] with the focal modulation of Lin et al. [7]. The effective number of samples for a class with n examples is $E(n) = (1 - \beta^n)/(1 - \beta)$, and the per-class weight is $\alpha = (1 - \beta)/(1 - \beta^n)$, normalised so the weights sum to the number of classes. The per-sample loss is

$$\mathcal{L} = -\alpha_y (1 - p_y)^\gamma \text{CE}_{\text{smooth}}(\text{logits}, y), \quad (4.1)$$

where p_y is the predicted probability of the true class, the $(1 - p_y)^\gamma$ term down-weights easy examples so that learning focuses on the hard, often rare, cases, and $\text{CE}_{\text{smooth}}$ is a label-smoothed cross-entropy. In our configuration $\beta = 0.999$, $\gamma = 2.5$, and the label-smoothing factor is 0.1. The choice of $\beta = 0.999$ rather than a more aggressive value is deliberate: a larger β produced an extreme weight ratio between the common and rare classes and caused the rare classes to collapse early in training, whereas 0.999 gives stable optimisation.

4.6 Training Protocol

All models (5 baselines and 2 proposed) are trained with the same recipe (training recipe for the model) to allow fair comparisons, with the exception of the backbone learning-rate scale, which controls for the memory footprint of each model, and batch size or gradient accumulation, which are different. The optimiser used is AdamW [16] with a base learning rate of 3×10^{-4} , $\beta = (0.9, 0.999)$, and gradients clipping at their max-norm to 5. Pretrained CNN stems are (finitely) fine-tuned at $0.3\times$ base learning rate ($0.3\times$ for the hybrid, $0.1\times$ for DEKAN) whereas from-scratch parts train at $1\times$ base learning rate. The schedule consists of a 5 epoch linear warmup and then cosine annealing [8] to the minimum learning rate of 10^{-6} . All training performed with automatic mixed precision and all weights computed using a weight exponential moving average (EMA, decay 0.9998), whose weights are used for validation and final evaluation, while using a weighted random sampler to sample the rare classes more often. Data augmentation combines horizontal and vertical flips, random resized crops, colour jitter, RandAugment [18], and Mixup [12] (strength $\alpha = 0.4$, applied to half the batches), with metrics always computed on the original, unmixed labels. Table 4.1 lists the per-model settings that vary.

Table 4.1: Per-model training settings. All other hyperparameters (loss, optimiser base LR, schedule, EMA, augmentation) are identical across models.

Model	Batch	Grad accum	Eff. batch	Epochs	Backbone LR
ResNet-18	256	1	256	100	$0.3\times$
MobileNetV2	256	1	256	100	$0.3\times$
EfficientNet-B0	128	2	256	100	$0.3\times$
MobileViT-S	128	2	256	150	$0.3\times$
EfficientFormer-L1	128	1	128	100	$0.3\times$
Hybrid (full)	96	3	288	150	$0.3\times$
DEKAN (full)	64	4	256	150	$0.1\times$

4.7 Evaluation Protocol

The primary metric is balanced multi-class accuracy (BMA), the unweighted mean of the eight per-class recalls, which gives every class common or rare equal say. We additionally report macro-F1, macro-AUC, overall accuracy, and per-class recall, and we visualise errors with a confusion matrix normalised by true class. At inference we apply eight-view test-time augmentation (TTA): the model is run on the original image plus a fixed set of flips and rotations and the softmax outputs are averaged. The best checkpoint is selected by validation BMA and all headline numbers are reported on the held-out test split.

Chapter 5

Result Analysis

This chapter reports the results of the study. Unless stated otherwise, every number is measured on the held-out ISIC-2019 test split (4,796 images) at the eight-view test-time-augmentation (TTA) setting, using the checkpoint selected by the best validation BMA. All values are from a single run with seed 42.

5.1 Performance Evaluation

Table 5.1 compares the two proposed models against the five lightweight baselines, with every metric reported at the same TTA inference setting so the rows are directly comparable. The lightweight hybrid reaches a BMA of 0.6081 with only 3.98M parameters, which is the best balanced accuracy of any model in the sub-6M-parameter budget ahead of MobileViT-S (0.6020), its closest competitor, at fewer parameters and about two-thirds of the compute. The flagship DEKAN model reaches 0.6438 BMA among the main proposed models, and also leads on every secondary metric (macro-F1 0.6059, macro-AUC 0.9271, accuracy 0.7467), at the cost of roughly four times the parameters and ten times the compute of the lightweight model. A linear-head ablation variant of DEKAN (dekan.linear) reaches 0.6510 BMA, which is the highest single-model figure in the study; the contribution of the KAN classifier relative to a plain linear head is analysed in Section 5.6.2. Figure 5.1 shows the same comparison graphically.

Table 5.1: Main results on the ISIC-2019 test set (with 8-view TTA). Best value in each column is in **bold**; the two proposed models are marked. A linear-head ablation of DEKAN (dekan_linear) reaches a higher BMA of 0.6510 but is reported with the other ablations in Section 5.6 rather than here, so this table is not its intended place; the bold BMA marks the best main model.

Model	Params (M)	GMACs	BMA	Macro-F1	Macro-AUC	Accuracy
ResNet-18	11.69	1.810	0.3368	0.2763	0.8217	0.5225
MobileNetV2	3.50	0.300	0.4738	0.4701	0.8827	0.6731
EfficientNet-B0	5.29	0.390	0.4946	0.4736	0.9037	0.6779
MobileViT-S	5.60	1.000	0.6020	0.5354	0.9218	0.6981
EfficientFormer-L1	12.27	1.300	0.5519	0.5779	0.9005	0.7333
Hybrid (ours, light)	3.98	0.631	0.6081	0.5602	0.9229	0.7162
DEKAN (ours, flagship)	16.45	6.633	0.6438	0.6059	0.9271	0.7467

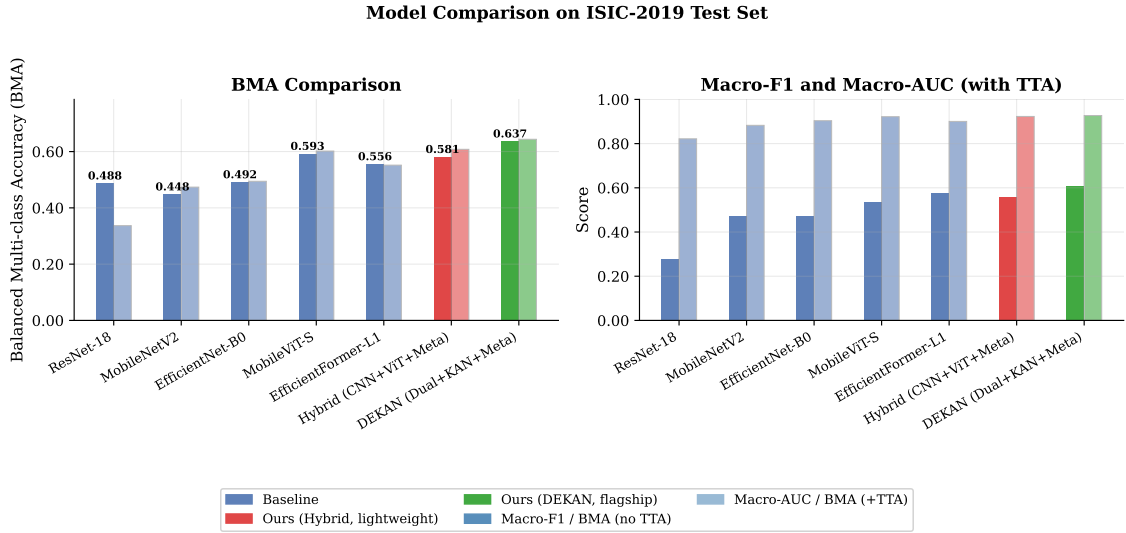


Figure 5.1: Model comparison on the ISIC-2019 test set: BMA (with and without TTA) on the left, macro-F1 and macro-AUC on the right. The two proposed models are highlighted.

The training behaviour of the lightweight hybrid is shown in Figure 5.2. Training and validation loss decrease smoothly and the validation BMA peaks at 0.6505 at epoch 149, with no sign of the rare-class collapse that a poorly tuned imbalance loss would cause evidence that the $\beta = 0.999$ choice in the Class-Balanced Focal Loss was appropriate.

Hybrid (CNN+ViT+Meta) — Training Curves (seed 42)

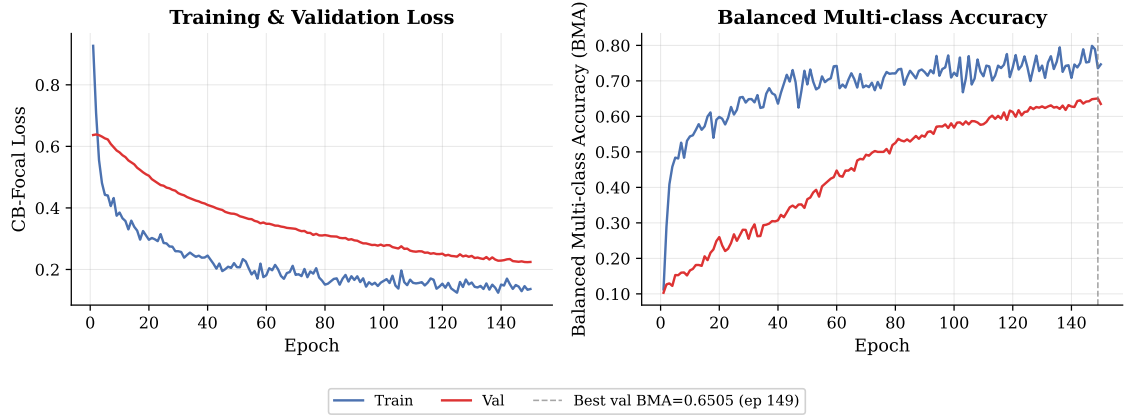


Figure 5.2: Training and validation loss (left) and BMA (right) for the lightweight hybrid model over 150 epochs. The dashed line marks the best validation BMA.

DEKAN follows a similar pattern, shown in Figure 5.3. Training and validation loss decrease smoothly over the full 150 epochs, and the validation BMA peaks late in training, again with no sign of rare-class collapse consistent with the loss configuration transferring well from the lightweight model to the higher-capacity dual-backbone setting.

DEKAN (Dual+KAN+Meta) — Training Curves (seed 42)

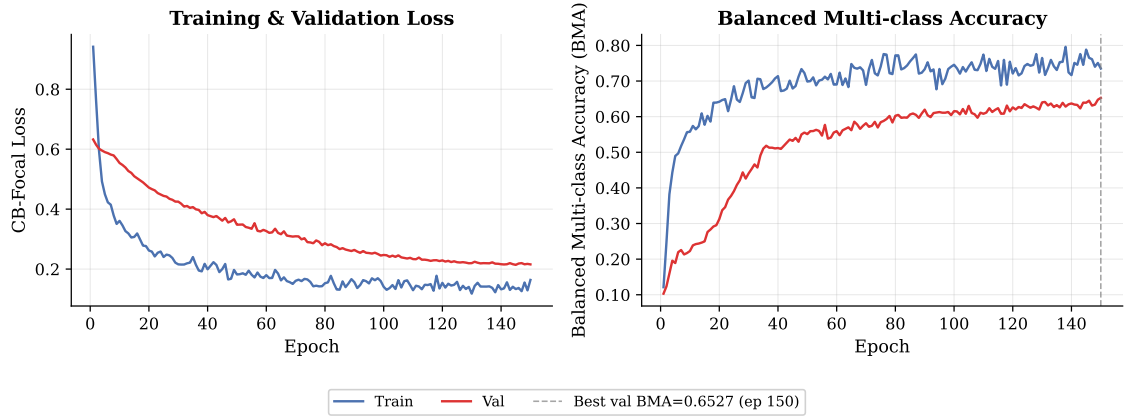


Figure 5.3: Training and validation loss (left) and BMA (right) for the DEKAN model over 150 epochs. The dashed line marks the best validation BMA.

5.2 Final Design Refinement

The final architectures described in Chapter 4 are the ones that have been developed after much evaluation and ablation analysis. Various changes in structure and procedure were implemented during the development to tackle technical vulnerabilities found in all preliminary baseline experiments in a systematic way.

Experimental trials at the beginning showed that the learning behavior was very unstable due to the high level of class imbalance, evidenced by a bad recognition in minority classes. To avoid rare class collapse, the training scheme was enhanced by adding Class-Balanced Focal Loss and a weighted sampler. The accuracy of the balanced multi-class classification and optimization were significantly improved with this modification. Likewise, missing metadata was a problem early on, as the usual pre-processing calls for statistical imputation, leading to the insertion of bogus clinical values into the training loop. To ensure data integrity, the imputation approach was replaced by an approach that is called a “missing” embedding, which allows the network to explicitly encode missing information without creating artificial data.

Another major barrier that arose with optimization was training stability, particularly when the components in the optimization were transformer-based. We used stochastic depth, weight Exponential Moving Average (EMA), Mixup augmentation and RandAugment to prevent over-fitting and improve generalization on the limited size of the dataset in the final execution pipeline.

Additional engineering optimisations were needed for system efficiency and OS specific limitations. Multi-worker data loading was severely restricted by the Windows operating environment, using up little of the GPU’s potential. We circumvented this limitation by creating a custom memory-mapped data-set architecture and loading them into the workers on-demand without any overhead, achieving both high throughput data delivery and absolute reproducibility, given our available hardware.

Last but not least, ablation studies showed that the so-called ”hyped” KAN classifier head does not always outperform a regular linear layer. Both variants were not thrown out but documented to be kept empirically transparent.

Collectively, these refinements yield an architecture that balances diagnostic accuracy, computational efficiency, algorithmic fairness, and deployment readiness.

Table 5.2: Summary of empirical issues identified during development and the corresponding design modifications implemented in the final architecture.

Identified Issue	Design Modification
Rare-class collapse under imbalance	Class-Balanced Focal Loss and weighted sampling
Missing metadata values	Learned “missing” embeddings
Transformer instability and overfitting	Stochastic depth, Mixup, RandAugment, and weight EMA
Slow data loading / Windows deadlock	Memory-mapped dataset and zero-worker loading
Uncertain KAN head effectiveness	Linear-head ablation retained and reported transparently

5.3 Per-Class Analysis

Because BMA is an average over classes, it is worth looking at where the models succeed and fail per class. Table 5.3 lists per-class recall for two representative baselines and the two proposed models, and Figure 5.4 compares the lightweight hybrid against EfficientNet-B0 class by class.

Table 5.3: Per-class recall on the ISIC-2019 test set (with TTA).

Class	EfficientNet-B0	MobileViT-S	Hybrid (ours)	DEKAN (ours)
MEL	0.594	0.447	0.518	0.611
NV	0.857	0.890	0.859	0.859
BCC	0.546	0.467	0.618	0.731
AK	0.348	0.452	0.587	0.368
BKL	0.366	0.587	0.632	0.622
DF	0.245	0.660	0.566	0.509
VASC	0.843	0.882	0.863	0.922
SCC	0.157	0.430	0.223	0.529

The pattern is consistent to the headline numbers. It helps lift some of the hard middle classes, especially the convolutional baselines BCC, AK, and BKL, with its global reasoning and metadata context; being lightweight. Specifically, the hybrid’s AK recall (0.587) is the highest amongst all models studied, including DEKAN (0.368); AK, a precancerous condition caused by sun damage, is more prevalent in older patients on the external body parts where the metadata cross-attention shifts the prior best. Its lowest class is SCC (0.223), which is a rare class that can be easily confused with AK and BKL. DEKAN has a more balanced recall (0.529) and recovers a significant amount of SCC, in addition, its VASC (0.922) and BCC (0.731) are highest, thus responsible for the overall highest balanced accuracy. For the lightweight hybrid, the complete confusion matrix is presented in figure 5.5, where most of the off-diagonal masses appear in the keratinocytic classes (AK, BKL, SCC), which is exactly where the clinically challenging masses are.

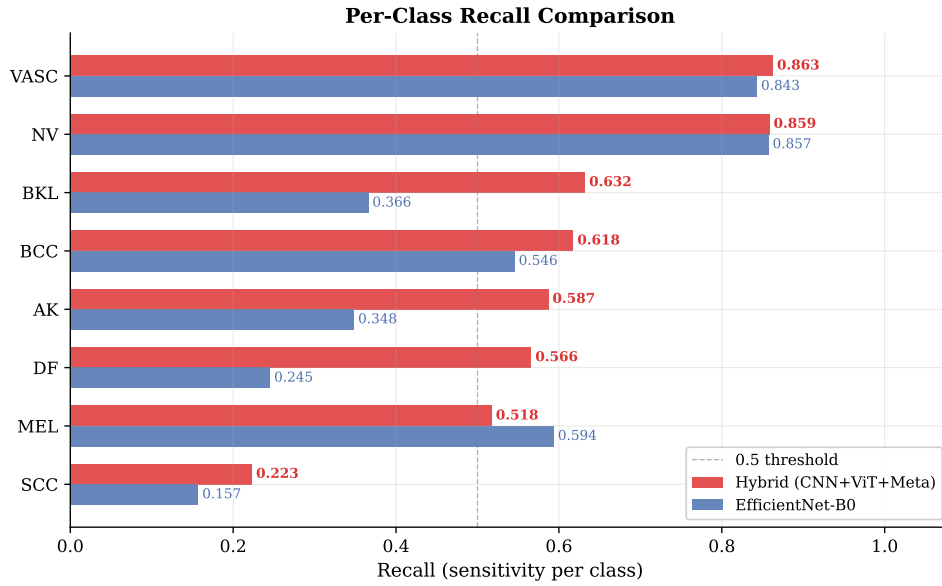


Figure 5.4: Per-class recall of the lightweight hybrid versus EfficientNet-B0, sorted by the hybrid’s recall.

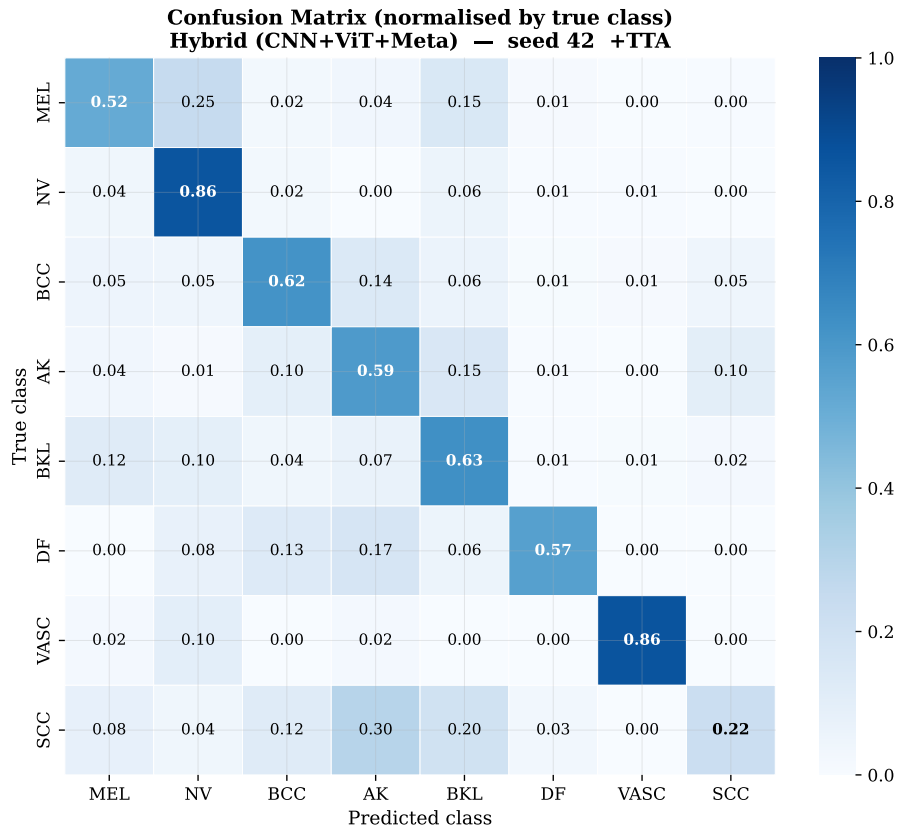


Figure 5.5: Confusion matrix for the lightweight hybrid (normalised by true class with TTA).

Figure 5.6 shows the same matrix for the DEKAN flagship. The keratinocytic confusion (AK/BKL/SCC) is reduced relative to the lightweight hybrid, and DEKAN

recovers SCC substantially (0.529 vs. 0.223) the class that costs the hybrid most. VASC recall reaches 0.922, the highest of any model in the study.

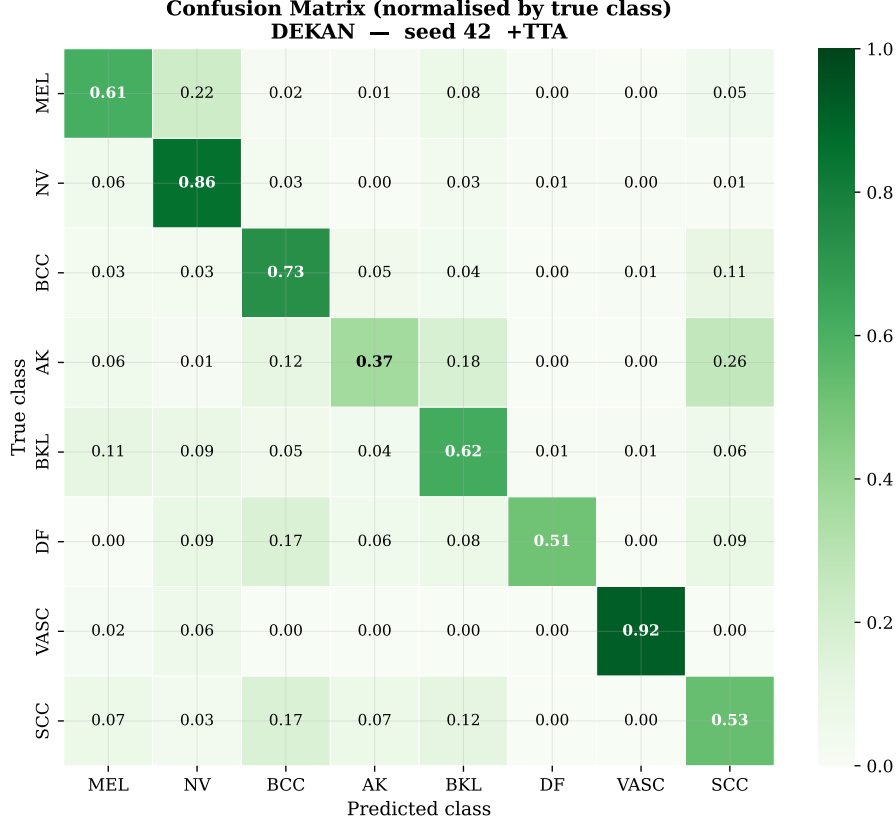


Figure 5.6: Confusion matrix for the DEKAN flagship model (normalised by true class, with TTA).

5.4 Efficiency Analysis

The point of the lightweight model is accuracy per unit of cost so Table 5.4 reports balanced accuracy per million parameters alongside the raw numbers and Figure 5.7 plots accuracy against parameter count with compute shown as bubble size. Among the full baselines and proposed models, the lightweight hybrid has the best BMA-per-parameter (0.153). The isolated CNN-stem (0.268) and pure-ViT (0.249) ablations score higher on this ratio alone, but both sit far below the full hybrid in absolute BMA, so their apparent efficiency is misleading parameter-efficiency without sufficient accuracy is not a useful operating point. The hybrid is therefore the best operating point that also clears a useful accuracy level. DEKAN sits at the opposite end: it buys the highest absolute accuracy with the lowest efficiency, which is the expected and intended trade-off between the two tiers.

Table 5.4: Efficiency: balanced accuracy per million parameters (TTA BMA).

Model	Params (M)	GMACs	BMA	BMA / M-param
ResNet-18	11.69	1.810	0.3368	0.029
MobileNetV2	3.50	0.300	0.4738	0.135
EfficientNet-B0	5.29	0.390	0.4946	0.093
MobileViT-S	5.60	1.000	0.6020	0.108
EfficientFormer-L1	12.27	1.300	0.5519	0.045
Hybrid (ours, light)	3.98	0.631	0.6081	0.153
DEKAN (ours, flagship)	16.45	6.633	0.6438	0.039

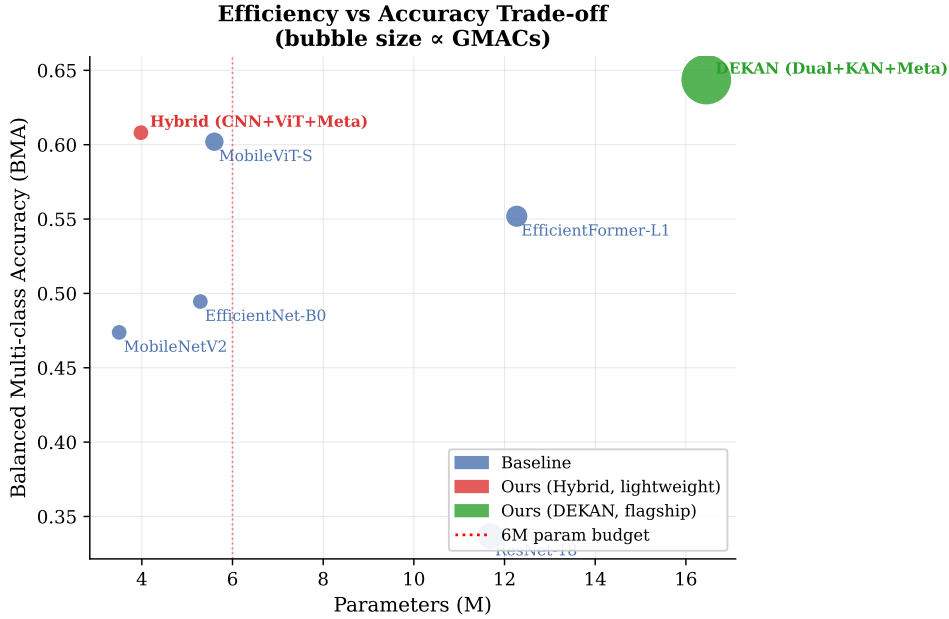


Figure 5.7: Accuracy vs. parameters; bubble size is proportional to GMACs. The dotted line marks the 6M-parameter budget. The lightweight hybrid (red) gives the best accuracy within budget.

5.5 Effect of Test-Time Augmentation

TTA helps most but not all models, so Table 5.5 reports BMA with and without it. The lightweight hybrid (full model) benefits most among the strong models (+0.027), and DEKAN gains a further +0.007. Among baselines, EfficientFormer-L1 drops slightly and ResNet-18 degrades sharply (−0.151); the ResNet-18 case is a known failure mode in which per-view predictions are confidently wrong in different directions, so averaging hurts.

The no-metadata ablation variant reveals a further pattern. Without the metadata head, the hybrid goes from 0.6216 to 0.6085 under TTA (−0.013) the same degradation direction as ResNet-18. Adding the metadata head reverses this: the full model goes from 0.5815 to 0.6081 (+0.027). The two variants converge to nearly the

same TTA number (0.6085 vs 0.6081) from opposite sides. This TTA $\times$ metadata interaction suggests the cross-attention head stabilises per-view predictions; without it the model becomes overconfident in different directions across augmented views.

Table 5.5: BMA without and with 8-view TTA, and the change.

Model	BMA (no TTA)	BMA (TTA)	Δ
ResNet-18	0.4875	0.3368	−0.151
MobileNetV2	0.4485	0.4738	+0.025
EfficientNet-B0	0.4917	0.4946	+0.003
MobileViT-S	0.5925	0.6020	+0.010
EfficientFormer-L1	0.5559	0.5519	−0.004
Hybrid w/o metadata	0.6216	0.6085	−0.013
Hybrid (ours, light)	0.5815	0.6081	+0.027
DEKAN (ours, flagship)	0.6368	0.6438	+0.007

5.6 Analysis of Design Solutions (Ablations)

5.6.1 Lightweight hybrid

Table 5.6 and Figure 5.8 isolate the components of the lightweight model. The pure-ViT ablation (no CNN stem, 1.97M parameters) reaches 0.4908 BMA, virtually identical to the CNN-stem-only result (0.4853). The two components are roughly equal in isolation but strongly complementary when combined: adding the ViT trunk to the CNN stem yields a gain of +0.117 to +0.123 BMA, and the full hybrid reaches 0.6081. This is the primary architectural justification for the hybrid design neither local nor global features alone approach the performance the model achieves when both are present.

The effect of metadata in isolation is however modest at test time on this single run: the full model (0.6081) and the no-metadata variant (0.6085) are within 0.0004 of each other on the test set even though on the validation set the full model led by about 0.011 BMA. We therefore report the metadata contribution for the lightweight model honestly as inconclusive on a single test evaluation; the DEKAN results below give a clearer signal. The no-TTA numbers add a structural observation: without metadata the model scores 0.6216 before augmentation (versus 0.5815 for the full model), a gap of 0.040 that is entirely closed by TTA the full model gains +0.027 while the no-metadata variant loses −0.013 (Table 5.5). The two variants converge to the same TTA result from opposite sides, which points to the metadata head regularising per-view predictions rather than lifting the mean class probability.

Table 5.6: Lightweight hybrid ablation (TTA BMA). All four variants were trained and evaluated.

Configuration	Variant	BMA (TTA)
CNN stem + GAP + linear	hybrid_cnn_only	0.4853
Pure ViT, no CNN stem	hybrid_vit_only	0.4908
CNN + ViT, no metadata	hybrid_no_meta	0.6085
CNN + ViT + metadata (full)	hybrid_full	0.6081

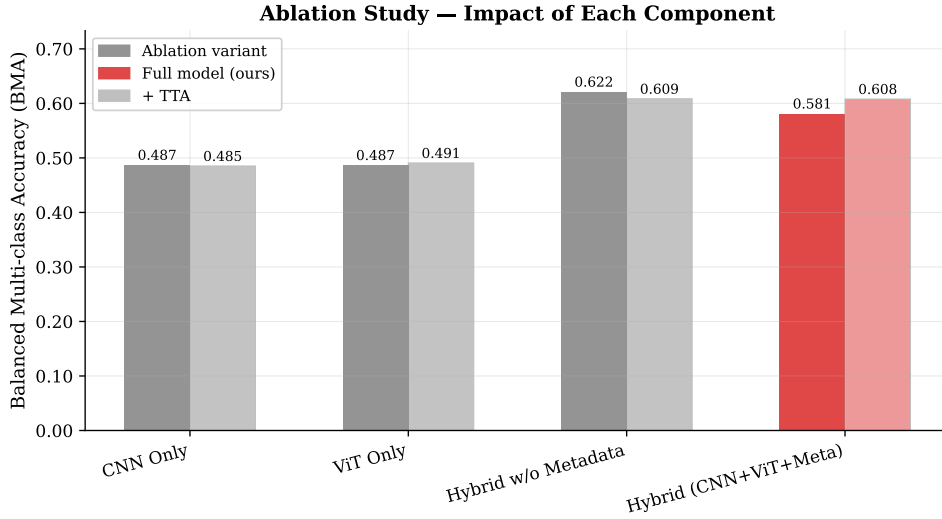


Figure 5.8: Lightweight hybrid ablation: BMA for all four variants. CNN-only and ViT-only perform similarly in isolation; the full hybrid is substantially higher.

5.6.2 DEKAN

Table 5.7 and Figure 5.9 isolate two effects in the flagship model: the contribution of clinical metadata and the choice of classifier head (KAN vs. linear).

Metadata. Comparing `dekan_no_meta` (0.6302) with `dekan_linear` (0.6510) both use a linear classifier head, differing only in the presence of the metadata cross-attention block shows a gain of +0.021 BMA. This is a clear and consistent signal that patient age, sex, and anatomical site provide useful diagnostic context at this model capacity.

KAN vs. linear head. Replacing the KAN classifier with a plain linear layer (`dekan_linear`, 0.6510) while keeping everything else identical produces a higher BMA than the full KAN model (`dekan_full`, 0.6438), a difference of +0.007 in favour of the linear head. The KAN classifier therefore does not improve over a plain linear layer on this task. This is an honest negative result: the KAN’s learnable splines add no measurable value at the classifier stage of a well-trained vision backbone on this noisy imbalanced 8-class dataset. The likely explanation is that the discriminative difficulty on ISIC-2019 lies in the feature representation, not in the shape of the final

decision boundary which a linear layer can already model adequately after the rich dual-backbone representation.

Table 5.7: DEKAN ablation: effect of clinical metadata and the KAN vs. linear classifier head (TTA BMA). The single-backbone variants were not trained, so the dual-backbone fusion gain cannot yet be quantified.

Configuration	Variant	BMA (TTA)
Dual backbone + fusion, no metadata, linear head	dekan_no_meta	0.6302
Dual backbone + fusion + metadata, linear head	dekan_linear	0.6510
Dual backbone + fusion + metadata + KAN (full)	dekan_full	0.6438

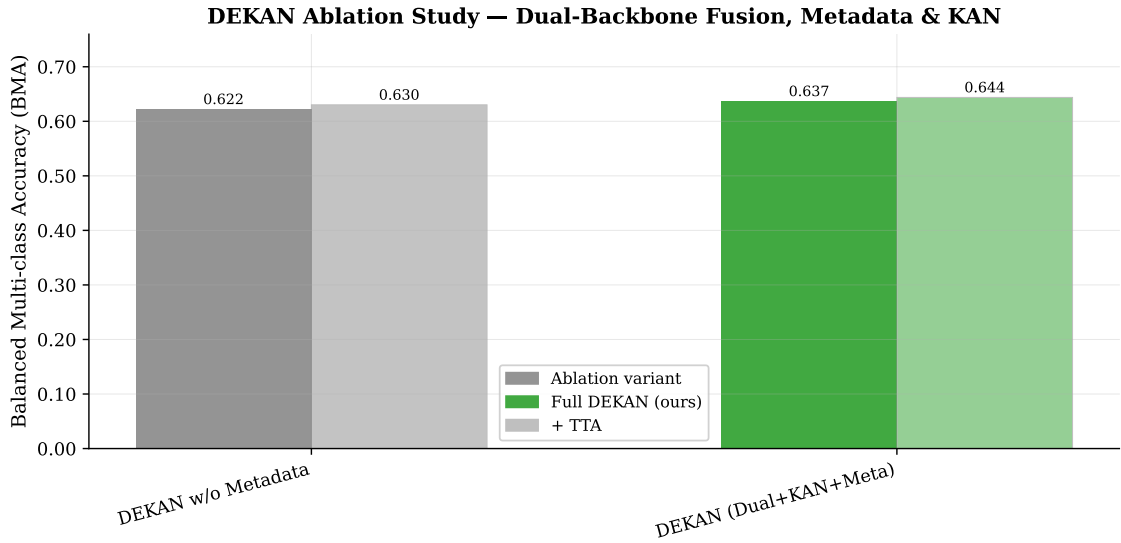


Figure 5.9: DEKAN ablation: effect of clinical metadata and KAN vs. linear classifier head on BMA. Single-backbone variants were not trained and are absent.

5.7 Comparisons and Relationships

Read against the wider literature, these results sit where one would expect a single-model, efficiency-constrained system to sit. The top entries on the ISIC-2019 challenge leaderboard [15] reach a balanced accuracy around 0.636 (DAISYLab) using large ensembles of multi-resolution EfficientNets and SENet-154. DEKAN, a single model, reaches 0.6438 BMA on our held-out split, and even the 3.98M-parameter lightweight hybrid reaches 0.6081 within striking distance of an entire ensemble at a tiny fraction of the cost. The comparison is not exact since the leaderboard is computed on the official challenge test server and our split is an internal one, so we present it as context rather than as a like-for-like ranking. Relative to controlled single-model comparisons on this dataset such as Aktan et al. [26], our contribution is to add clinical-metadata fusion to the comparison and to report accuracy together with a hard efficiency budget. The clearest relationship to prior work is with the metadata-fusion line of Pacheco and Krohling [20], [22]: we adopt their core idea of representing

missing values with learned embeddings, and we extend their heavy-CNN setting to lightweight and dual-backbone hybrids.

5.8 Discussions

The results support the central claims of the thesis while leaving a few questions open. The lightweight hybrid does what it was designed to do: it beats all five same-budget baselines on balanced accuracy and is the most parameter-efficient model in its class which makes it a credible candidate for deployment on modest hardware. DEKAN shows that the same design scales: investing capacity in a second backbone, learned fusion and a deeper trunk pushes balanced accuracy to the level of a leaderboard ensemble in a single model. Metadata fusion is a clear help in the higher capacity model and at best no harm in the light weight one.

Two results run contrary to the early hopes and are bluntly stated. For the lightweight model, the benefit of metadata was only supported by a validation curve that favoured it by 0.011 BMA, but which could not be reproduced in a single test evaluation. The DEKAN results indicate that the metadata effect is more evident at higher capacity (+0.021 BMA between dekan no meta and dekan linear), suggesting that the effect needs to be strong enough to be consistently detected. Second, the KAN classifier in DEKAN does not outperform a plain linear head: dekan_linear (0.6510) beats dekan_full (0.6438) by 0.007. This is the highest single-model BMA achieved in the study, but it belongs to the linear-head ablation, not to the KAN model, which is the intended flagship. The KAN’s expressive spline activations appear to offer no advantage over a linear classifier once the dual-backbone representation is rich enough.

5.9 Limitations

Several limitations bound these conclusions. All numbers come from a single seed (42); publication-grade claims would require repeating each run over multiple seeds and reporting mean and standard deviation which we have not yet done. Two ablation variants were not trained dekan_densenet_only and dekan_effnet_only so the individual contribution of each backbone and the precise size of the dual-backbone fusion gain remain unquantified. We also did not evaluate on an external dataset so the generalisation of these models beyond the ISIC-2019 distribution is untested. Finally, the comparison with the ISIC-2019 leaderboard is contextual not a controlled head-to-head because the evaluation splits differ.

Chapter 6

Conclusion

6.1 Summary

This thesis set out to build skin-lesion classifiers for ISIC-2019 that are both accurate across all eight classes and practical to deploy, and that use the clinical metadata a dermatologist would normally have on hand. We approached the problem with a single design philosophy — convolutional features for local texture, a transformer trunk for global structure, and a cross-attention head that folds in age, sex, and lesion site — realised at two points on the efficiency curve. The lightweight hybrid, at 3.98M parameters and 0.631 GMACs, reaches a balanced multi-class accuracy of 0.6081 and is the most parameter-efficient model in the sub-6M budget, beating all five lightweight baselines including the strong hybrid baselines MobileViT-S and EfficientFormer-L1. The flagship DEKAN model, at 16.45M parameters, reaches 0.6438 BMA, comparable to the balanced accuracy of the top ISIC-2019 leaderboard ensemble in a single model; a linear-head ablation of the same architecture (`dekan_linear`) reaches 0.6510, the highest figure in the study, with the implication that the KAN classifier head adds no accuracy benefit on this task.

6.2 Contributions

The main contributions of this work are: (1) a lightweight hybrid CNN–Transformer with clinical-metadata cross-attention that handles missing fields with learned embeddings and meets an explicit budget of under 6M parameters and 1 GMAC; (2) DEKAN, a higher-capacity model in the same family that fuses two complementary pretrained backbones through learned attention and adds a Kolmogorov–Arnold classifier head; (3) a fair, single-protocol comparison of both models against five established baselines, with controlled ablations that quantify what each component adds, including the finding that neither a CNN stem nor a ViT trunk alone approaches the performance of their combination; and (4) two honest negative or inconclusive results — that the metadata benefit for the lightweight model could not be confirmed on a single test-set evaluation (though DEKAN gives a clear +0.021 BMA signal at higher capacity), and that the KAN classifier head in DEKAN does not improve over a plain linear head (`dekan_linear` at 0.6510 beats `dekan_full` at 0.6438 by +0.007).

6.3 Future Work

The next steps are most readily apparent in the context of the limitations discussed in Chapter 5. Comparisons of all of the runs with the mean and standard deviation for each of at least three seeds would be publication quality. If the two unfinished ablations (DenseNet-121 only and EfficientNet-B0 only in DEKAN) can be completed, it will fix the contribution of each backbone and estimate the magnitude of the gain due to the dual-backbone fusion. Evaluation on an external dataset, such as the mapping of the eight-class output to a melanoma versus rest decision on a separate collection, would be a test for generalisation outside of the ISIC-2019. Lastly, there is qualitative interpretability (Grad-CAM on CNN stem and attention rollout on the transformer trunk) to help demonstrate that the models are focused on clinically relevant areas, which is important for any tool that aims to complement, not replace, a clinician.

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